

Trichoptera and Plecoptera of the Seybouse River, northeast Algeria: Distribution, phenology and new records

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Abstract

The stoneflies and caddisflies of North Africa are still poorly known as vast areas of Algeria have yet to be investigated. A survey of the macroinvertebrates of the Seybouse River, northeast Algeria, was carried out from July 2014 to December 2016. Three species of stoneflies (*Capnopsis schilleri*, *Capnioneura petitpierreae*, and *Tyrrhenoleuctra tangerina*) and five taxa of caddisflies (*Mesophylax aspersus*, *Hydropsyche maroccana*, *H. resmineda*, *H. artax/lobata*, and *H. gr. pellucidula*) were identified. All taxa are new records to the Seybouse River and seven of them are new to northeastern Algeria. A multivariate analysis indicated that the Hydropsychidae exhibited a clear longitudinal gradient along the Seybouse River while *Mesophylax aspersus* seemed adapted to species-poor, intermittent streams. Further investigations of the stoneflies and caddisflies may inform conservation efforts and will prove useful to monitor the Seybouse River and similarly threatened North African rivers and streams.

Key words: Stoneflies, caddisflies, biodiversity, distribution, Maghreb, North Africa

Introduction

The first check-list of Algerian stoneflies was provided by Lestage (1925) based in part on Selys' collection (Klapálek 1923) and surveys conducted by Kolbe (1895) and Gauthier (1928). Further works by Aubert (1956) and Berthélemy (1973) consolidated our knowledge of the Plecoptera of Algeria. The last three decades have witnessed a series of studies which have considerably improved our understanding of the taxonomy and distribution of this group (Gagneur & Aliane 1991; Lounaci & Vinçon 2005, Vinçon & Pardo 2006; Yasri *et al.* 2013; Yasri-Cheboubi *et al.* 2013, 2016). However, vast areas of Algeria remain unexplored and it is expected that further investigations would advance our knowledge further.

Studies of the caddisflies of Algeria were pioneered by McLachlan (1880) and Morton (1896, 1898) in the late 19th Century. These initial records were followed in the early 20th Century by investigations by Ulmer (1905), Navás (1917, 1925, 1932, 1936), and Lestage (1925). A few decades later, more species were added by Vaillant (1953a,

1953b, 1954a, 1954b) and Schmid (1957). As happened with stoneflies, our knowledge of the taxonomy and distribution of caddisflies of Algeria improved considerably over the last four decades (Malicky & Lounaci 1987; Arab *et al.* 2004; Sekhi *et al.* 2016, 2019; Dambri *et al.* 2020). These investigations have revealed an extended distribution of several species thought previously to be confined to other parts of the Mediterranean Region (Malicky & Lounaci 1987).

Although the macroinvertebrates of the Seybouse River have been the focus of a series of faunistic and ecological investigations carried out by the Laboratoire de Recherche des Zones humides (Odonata: Yalles-Satha & Samraoui 2017; Bouhala *et al.* 2019; Samraoui *et al.* 2019; Hemiptera: Annani *et al.* 2012; Benslimane *et al.* 2019; Chironomidae: Chaib *et al.* 2013; Simuliidae: Cherairia *et al.* 2014; Ephemeroptera: Bouhala *et al.* 2019, 2020a, 2020b), the stoneflies and the caddisflies of this important North African river have, so far, not been tackled.

Methods

Study area

The Seybouse River, one of the most important Algerian hydrosystem, is a Mediterranean river which originates in the Semi-arid Hauts Plateaux before flowing northward to reach the Mediterranean Sea at Annaba. It measures 134.7 km and drains a basin spread over 6570 km². The confluence of its main tributaries, Wadi Cherf and Wadi Bouhamdane, is located near Guelma. The climate is typically Mediterranean, divided into a rainy season from October to May and a dry season from June to September. Rainfall ranges from 350 mm/year upstream to 610 mm/year downstream (Yalles-Satha & Samraoui 2017). In contrast to the downstream region of Wadi Seybouse, the headwaters lack industrial plants in their vicinity but points of untreated sewage discharge are often present near scattered human settlements. Watercourses are commonly used as a source of water for grazing livestock (Bouhala *et al.* 2020b).

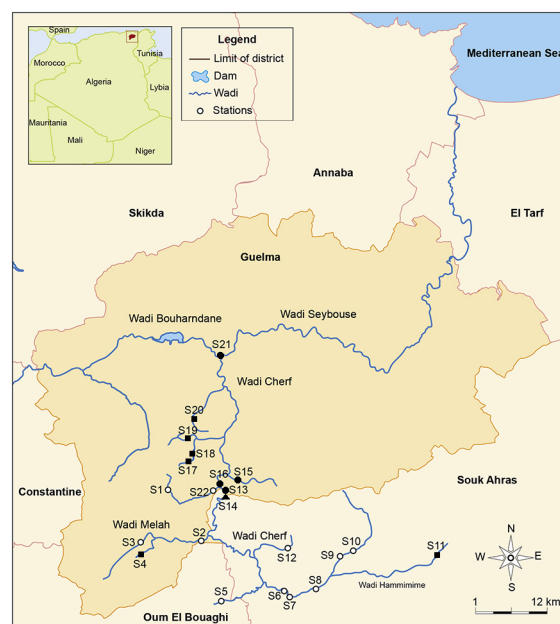


FIGURE 1: Map of Algeria with location of sampling stations (S1–S22) at Wadi Cherf, a tributary of the Seybouse River. The station codes are provided in Table 1. Dark circles = *Hydropsyche* spp., Dark squares = *Mesophylax aspersus* (Rambur 1842), Dark triangle = *Hydropsyche* sp. + *Mesophylax aspersus*).

Sampling

Monthly sampling was performed from July 2014 to June 2017, but samples from only July 2014 to December 2016 will be considered here. Records of adult and larvae of Plecoptera and Trichoptera were noted at 22 stations

(S1–S22), selected along the upper reaches of the Seybouse River, and distributed between altitudes 250–860 m (Fig. 1, Table 1). Samples of macroinvertebrates were collected using a dipnet (500 µm mesh size, 35 cm diameter) and were fixed in 5% formaldehyde. In each locality, an area of 100 m² was kick-sampled by walking slowly and repeatedly across all microhabitats for ten minutes. Plecoptera were identified by J.M. Tierno de Figueroa and G. Vinçon; Trichoptera were identified by A. Ruiz Garcia.

TABLE 1: List of sampled localities with GPS coordinates.

Code	Name	Latitude (N)	Longitude (E)	Altitude (m)
S1	Beldjoudi	36° 13' 23"	7° 12' 52"	822
S2	Ben Mheni	36° 9' 2"	7° 16' 50"	667
S3	O. El Maleh	36° 8' 53"	7° 8' 38"	748
S4	Aïn Arko	36° 7' 35"	7° 9' 17"	750
S5	O. Dahmane	36° 3' 11"	7° 19' 40"	752
S6	Sedrata Dam	36° 3' 34"	7° 27' 19"	746
S7	Sedrata Bridge II	36° 3' 8"	7° 27' 58"	760
S8	Sedrata Bridge I	36° 4' 28"	7° 29' 39"	765
S9	O. Krab	36° 7' 12"	7° 32' 46"	784
S10	Krab Bridge	36° 7' 55"	7° 34' 16"	800
S11	Tiffrech Bridge	36° 8' 28"	7° 45' 16"	860
S12	O. Nil	36° 8' 23"	7° 26' 45"	864
S13	Dbabcha	36° 13' 0"	7° 19' 5"	610
S14	O. Aar	36° 13' 34"	7° 19' 11"	617
S15	O. Cheniour	36° 20' 30"	7° 23' 39"	602
S16	Aïn Makhoulf Bridge	36° 14' 23"	7° 18' 41"	590
S17	Aazlat Hlima	36° 16' 11"	7° 17' 1"	770
S18	Chaabet Bounab	36° 17' 5"	7° 17' 4"	805
S19	O. Boufrais	36° 18' 57"	7° 16' 20"	612
S20	Selaoua	36° 20' 54"	7° 16' 2"	589
S21	Medjez Amar	36° 26' 32"	7° 18' 41"	245
S22	Aïn Makhoulf Dam	36° 13' 34"	7° 17' 48"	643

Statistical analysis

A Principal Component Analysis (PCA) was performed with a matrix sites/Trichoptera species. Data analysis was based on records of 2015 and 2016. The data matrix concerning the Plecoptera was too sparse, with only three sites involved, and thus was not included in any analysis. The statistical analysis was accomplished using R (R Development Core Team 2020).

Plecoptera

Family Capniidae

Capnopsis schilleri (Rostock 1892), **first record for the Seybouse River and northeastern Algeria** (Table 2).

This species was recorded from only one site (S12) in March.

Oued Nil (S12): 12.iii.2015, 3 larvae.

This Palearctic species, previously recorded in Algeria in the Djurdjura Mountains (Lounaci *et al.* 2000; Lounaci & Vinçon 2005), is widely distributed in Europe and the Maghreb (Zwick 1984). The rheophilic larvae are detritivorous (López-Rodríguez & Tierno de Figueroa 2008) and are found on both sandy and stony substrates in mountain streams (Gallardo-Mayenco 1990; Brittain & Saltveit 1996) with altitudes ranging from sea level (<200 m) to montane heights (1000–1900 m) (Graf *et al.* 2009). In the Seybouse River and elsewhere in Algeria, the species has been found at modest altitudes (400–600 m) (B.S. unpublished).

TABLE 2: Checklist of Plecoptera and Trichoptera of the Seybouse River

Plecoptera		
Capniidae		<i>Capnopsis schilleri</i> (Rostock 1892)
		<i>Capnioneura petitpierreae</i> Aubert 1961
Leuctridae		<i>Tyrrhenoleuctra tangerina</i> (Navás 1922)
Trichoptera		
Hydropsychidae		<i>Hydropsyche maroccana</i> Navás 1936
		<i>Hydropsyche resmineda</i> Malicky 1977
		<i>Hydropsyche artax/lobata</i>
		<i>Hydropsyche pellucidula</i> Species Group
Limnephilidae		<i>Mesophylax aspersus</i> (Rambur 1842)

Capnioneura petitpierreae Aubert 1961, **first record for the Seybouse River.**

This species was also recorded from a single site (S20) in March.

Selaoua (S20): 13.iii.2015, 2 larvae.

This Ibero-Maghrebian species is widely distributed in the Maghreb but rare in southern Spain (Aubert 1961; Tierno de Figueroa *et al.* 2003; Errochdi *et al.* 2014). The detritivorous larvae are eurytopic (Azzouz & Sánchez-Ortega 2000) and occupy streams with an altitudinal range from 10 to 2000 m (Errochdi *et al.* 2014). It has been reported from western Algeria (Gagneur & Aliane 1991).

Family Leuctridae

Tyrrhenoleuctra tangerina (Navás 1922), **first record for the Seybouse River.**

This species seems much more common than *Capnopsis schilleri* and *Capnioneura petitpierreae* and was recorded from three sites S12, S19, and S20 from February to April.

Oued Nil (S12): 12.iii.2015, 2 larvae; 23.iv.2015, 1 larva; 18.ii.2016, 2 larvae; 29.iii.2016, 1 larva. Boufrais (S19): 13.iii.2015, 3 larvae; 19.ii.2016, 1 larva; 29.iii.2016, 1 larva; 11.iii.2017, 1 larva. Selaoua (S20): 13.iii.2015, 1 larva.

The genus *Tyrrhenoleuctra* includes a complex of cryptic species that have been the focus of concerted molecular investigations that separated *T. tangerina*, an Ibero-Maghrebian species from *T. minuta* recorded in southern Spain (Fochetti *et al.* 2009; Fochetti & Tierno de Figueroa 2009; Tierno de Figueroa & Fochetti 2014). In Algeria, the larvae inhabit temporary streams with an altitudinal gradient ranging from 200 to 900 m (B.S. unpublished).

Trichoptera

Family Hydropsychidae

Hydropsyche maroccana Navás 1936, **first record for the Seybouse River and northeastern Algeria.**

This species (Fig. 2a) was collected at five sites (Fig. 1): Dbebcha (S13), Oued Aar (S14), Oued Cheniour (S15), Aïn Makhoulf Bridge (S16), and Medjez Amar (S21). Larvae were found all year long while pupae were found from February to July.

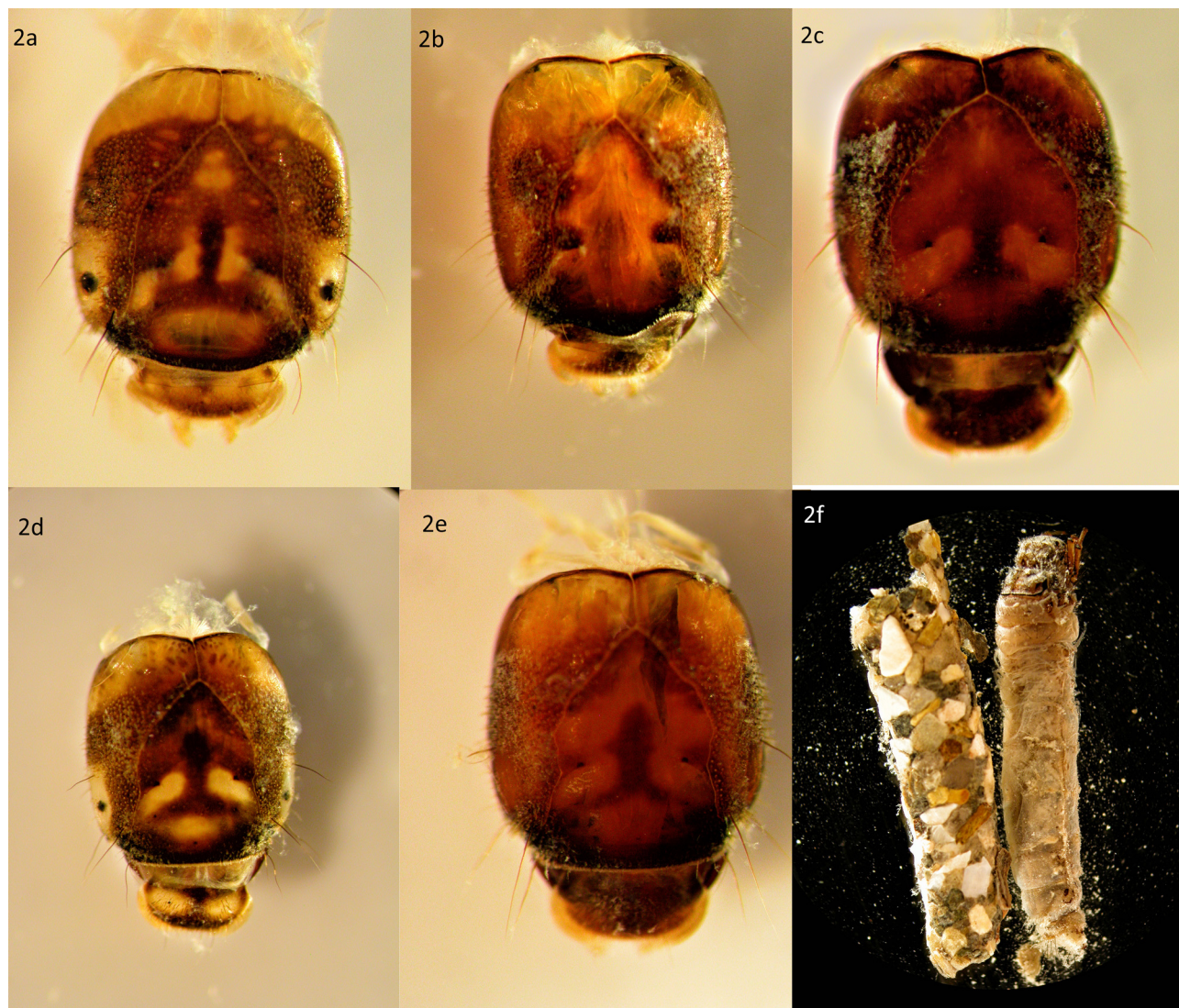


FIGURE 2: Trichoptera larvae. 2a–2e, *Hydropsyche* cephalic capsules of the Seybouse River, northeast Algeria: 2a, *H. maroccana* Navás 1936; 2b, *H. resmineda* Malicky 1977; 2c, *H. artax/lobata*; 2d, *H. gr. pellucidula*, form 1; 2e, *H. gr. pellucidula*, form 2. 2f, *Mesophylax aspersus* (Rambur 1842) of the Seybouse River, northeast Algeria, larva and case.

Dbebcha (S13): 04.vi.2016, 1 pupa; 07.x.2016, 1 larva; 12.xii.2016, 1 larva. Oued Aar (S14): 07.vi.2015, 1 larva; 22.vii.2015, 3 larvae; 14.xi.2015, 2 larvae; 01.i.2016, 1 larva; 19.ii.2016, 3 larvae; 23.iv.2016, 2 larvae; 09.v.2016, 2 larvae. Oued Cheniour (S15): 25.xii.2014, 1 larva; 07.vi.2015, 1 larva; 11.viii.2015, 1 larva; 13.xii.2015, 2 larvae; 19.ii.2016, 1 pupa; 29.iii.2016, 2 larvae; 23.iv.2016, 4 larvae; 09.v.2016, 4 larvae; 04.vi.2016, 3 larvae. Aïn Makhoulf Bridge (16): 30.vii.2014, 1 larva, 23.viii.2014, 4 larvae; 28.ix.2014, 1 larva; 27.x.2014, 1 larva; 25.xii.2014, 2 larvae; 26.ii.2015, 1 larva; 21.iv.2015, 2 larvae; 07.vi.2015, 3 larvae; 24.vii.2015, 3 larvae; 08.viii.2015, 1 larva; 27.x.2015, 6 larvae; 13.xii.2015, 1 larva; 29.iii.2016, 1 larva, 1 pupa; 23.iv.2016, 3 larvae; 04.vi.2016, 1 larva; 18.viii.2016, 2 larvae; 07.X.2016, 1 larva; 01.xi.2016, 3 larvae. Medjez Amar (S21): 19.ii.2016, 2 larvae; 29.iii.2016, 18 larvae, 1 pupa; 23.iv.2016, 14 larvae, 6 pupae; 09.v.2016, 9 larvae, 1 pupa; 04.vi.2016, 2 larvae; 15.vii.2016, 2 larvae; 16.viii.2016, 2 larvae; 28.ix.2016, 2 larvae; 07.x.2016, 10 larvae; 01.xi.2016, 4 larvae.

Hydropsyche maroccana is endemic to the Maghreb and the Canary Islands (Tobias & Tobias 2008; Neu *et al.* 2018), and has been found in Algeria previously only in the Djurdjura Mountains and the Chelif River (Malicky & Lounaci 1987; Arab *et al.* 2004).

Hydropsyche resmineda Malicky 1977, **first record for the Seybouse River and northeastern Algeria.**

This species (Fig. 2b) was collected at six sites (Fig. 1): Krab Bridge (S10), Dbebcha (S13), Oued Aar (S14), Oued Cheniour (S15), Aïn Makhoulf Bridge (S16), and Medjez Amar (S21). Larvae were found all year long while pupae were recorded from February to September.

Krab Bridge (S10): 22.vii.2015, 4 larvae. Dbebcha (S13): 30.vii.2014, 3 larvae; 13.viii.2014, 2 larvae; 27.x.2014, 1 larva; 25.xii.2014, 1 larva; 21.iv.2015, 7 larvae; 7.vi.2015, 10 larvae; 22.vii.2015, 3 larvae; 8.viii.2015, 2 larvae; 16.ix.2015, 1 larva; 27.x.2015, 6 larvae; 14.xi.2015, 4 larvae; 29.iii.2016, 7 larvae; 23.iv.2016, 4 larvae; 04.vi.2016, 2 larvae; 16.vii.2016, 4 larvae, 1 pupa; 16.viii.2016, 8 larvae; 07.x.2016, 2 larvae. Oued Aar (S14): 11.v.2015, 2 larvae; 07.vi.2015, 1 larva; 22.vii.2015, 5 larvae, 1 pupa; 15.ix.2015, 3 larvae; 27.x.2015, 3 larvae; 14.xi.2015, 1 larva; 13.xii.2015, 3 larvae; 01.i.2016, 4 larvae; 19.ii.2016, 4 larvae; 29.iii.2016, 2 larvae, 1 pupa; 23.iv.2016, 2 larvae, 1 pupa; 09.v.2016, 2 larvae; 04.vi.2016, 1 pupa. Oued Cheniour (S15): 27.x.2015, 2 larvae; 13.xii.2015, 1 larva; 19.ii.2016, 2 pupae; 29.iii.2016, 1 larva. Aïn Makhoulf Bridge (S16): 30.vii.2014, 12 larvae; 23.viii.2014, 14 larvae, 1 pupa; 28.ix.2014, 11 larvae, 1 pupa; 25.xii.2014, 1 larva; 16.iii.2015, 5 larvae; 21.iv.2015, 4 larvae; 11.v.2015, 1 larva, 2 pupae; 07.vi.2015, 9 larvae; 08.viii.2015, 3 larvae; 27.x.2015, 17 larvae; 14.xi.2015, 7 larvae; 01.i.2016, 2 larvae; 29.iii.2016, 4 larvae, 3 pupae; 23.iv.2016, 3 pupae; 16.viii.2016, 3 larvae; 07.x.2016, 5 larvae; 01.xi.2016, 4 larvae; 12.xii.2016, 3 larvae. Medjez Amar (S21): 29.xi.2014, 1 larva; 21.iv.2015, 4 larvae, 1 pupa; 12.v.2015, 1 larva; 12.vii.2015, 6 larvae, 1 pupa; 08.viii.2015, 3 larvae, 1 pupa; 16.ix.2015, 3 larvae; 14.xi.2015, 2 larvae; 13.xii.2015, 5 larvae; 19.ii.2016, 3 larvae; 29.iii.2016, 5 larvae, 2 pupae; 23.iv.2016, 2 larvae; 04.vi.2016, 1 larva; 28.ix.2016, 1 larva; 07.x.2016, 1 larva.

This Maghrebian endemic has been recorded in Algeria (Malicky 1983) and the rest of the Maghreb (Tobias & Tobias 2008; Neu *et al.* 2018).

Hydropsyche artax/lobata, **first record for the Seybouse River and northeast Algeria.**

This taxon (Fig. 2c) was collected at five sites (Fig. 1): Dbebcha (S13), Oued Aar (S14), Oued Cheniour (S15), Aïn Makhoulf Bridge (S16), and Medjez Amar (S21). Larvae were found all year long while pupae were recorded in only May and July.

Dbebcha (S13): 27.x.2014, 1 larva; 21.iv.2015, 2 larvae; 19.ii.2016, 1 larva; 29.iii.2016, 2 larvae; 09.v.2016, 1 larva; 12.xii.2016, 3 larvae. Oued Aar (S14): 19.i.2015, 1 larva; 11.v.2015, 1 larva; 07.vi.2015, 8 larvae; 22.vii.2015, 1 larva, 2 pupae; 15.ix.2015, 1 larva; 27.x.2015, 2 larvae; 01.i.2016, 1 larva; 19.ii.2016, 3 larvae; 23.iv.2016, 1 larva. Oued Cheniour (S15): 25.xii.2014, 2 larvae; 11.viii.2015, 1 larva; 01.i.2016, 1 larva; 29.iii.2016, 3 larvae; 23.iv.2016, 1 larva; 09.v.2016, 1 pupa. Aïn Makhoulf Bridge (S16): 29.xi.2014, 1 larva; 25.xii.2014, 1 larva; 13.xii.2015, 1 larva; Medjez Amar (S21): 17.x.2014, 1 larva; 29.xi.2014, 3 larvae; 12.v.2015, 22 larvae; 11.vi.2015, 8 larvae; 22.vii.2015, 1 larva, 1 pupa; 08.viii.2015, 6 larvae; 16.ix.2015, 2 larvae; 13.xii.2015, 2 larvae; 29.iii.2016, 2 larvae; 14.xi.2015, 3 larvae; 13.xii.2015, 1 larva; 23.iv.2016, 2 larvae; 09.v.2016, 2 larvae.

The taxonomic status of the larvae of *H. lobata* McLachlan 1884 and *H. artax* Malicky & Lounaci 1987 is somewhat contentious. According to Neu *et al.* (1998) and Malicky (1999), both species are valid Maghrebian species, whereas Botosaneanu (1999) synonymized *H. artax* with *H. lobata*. However, his view was not shared by Tobias & Tobias (2008) who cited both species from Algeria.

Hydropsyche pellucidula Species Group

This taxon (Fig. 2d, 2e) was collected at four sites (Fig. 1): Dbebcha (S13), Oued Aar (S14), Oued Cheniour (S15), and Aïn Makhoulf Bridge (S16). Larvae were found all year long while pupae were recorded in only April and May.

Dbabcha (S13): 29.iii.2016, 1 larva. Oued Aar (S14): 07.vi.2015, 3 larvae; 08.viii.2015, 1 larva; 01.i.2016, 1 larva; 19.ii.2016, 1 larva. Oued Cheniour (S15): 11.viii.2015, 1 larva; 16.ix.2015, 2 larvae; 14.xi.2015, 2 larvae; 13.xii.2015, 1 larva; 01.i.2016, 1 larva; 29.iii.2016, 1 larva; 23.iv.2016, 2 larvae, 3 pupae; 09.v.2016, 1 pupa; 15.vii.2016, 1 larva; 16.viii.2016, 3 larvae. Aïn Makhoul Bridge (S16): 21.iv.2015, 1 larva.

The *Hydropsyche pellucidula* Species Group includes some species that could occur in northeastern Algeria, including *H. punica* Malicky 1981, *H. pellucidula* (Curtis 1834), *H. pandiel* Malicky 2014 (larva unknown), and *H. iberomaroccana* González & Malicky 1999. The known larvae of these *Hydropsyche* spp. all bear a distinctive cephalic capsule in the last instar (Fig. 2). However, the genus is poorly known in the Maghreb (Dakki 1978; Malicky & Lounaci 1987) and identification needs to be supported by mature pupae and/or adults. A synoptic key of the *Hydropsyche* taxa recorded in the Seybouse River, based on the nomenclature of Zamora-Muñoz *et al.* (1995) is provided in Table 3.

TABLE 3. Synoptic key for the *Hydropsyche* species recorded in the Seybouse River based on frontoclypeal apotome characteristics of the larvae, according to the nomenclature used by Zamora-Muñoz *et al.* (1995).

Frontoclypeal Apotome	<i>Hydropsyche</i> <i>artax/lobata</i> (Fig. 2c)	<i>Hydropsyche</i> <i>pellucidula</i> Species Group (Figs 2d, 2e)	<i>Hydropsyche</i> <i>maroccana</i> (Fig. 2a)	<i>Hydropsyche</i> <i>resmineda</i> (Fig. 2b)
Anterior edge of apotome	Straight	Straight	Convex	Strongly convex
Shape of apotome	Wide	Narrow	Narrow	Narrow
Symmetry of anterior edge with respect to longitudinal axis	Symmetrical	Symmetrical	Symmetrical	Asymmetrical
Light spots distribution on apotome	Two elongated light spots on epistomal sulcus, sometimes fused with light oral spot, when present	Two symmetrical light lateral spots on epistomal sulcus and one ovoid oral light spot present	Oval light oral spot; two symmetrical light lateral spots each extended by longitudinal light spot; small light aboral spot at posterior angle	Overall, little dark coloration; two black, ovoid, horizontal and symmetrical lateral spots present at pretentorinae level

Family Limnephilidae

Mesophylax aspersus (Rambur 1842), **first record for the Seybouse River and northeast Algeria.**

This species (Fig. 2f) was collected at seven sites (Fig. 1): Aïn Arko (S4), Tiffech Bridge (S11), Oued Aar (S14), Aazelet Hlima (S17), Chaabet Bounab (S18), Boufrais (S19), and Selaoua (S20). Larvae were found from December to April while pupae were recorded in only April and May.

Aïn Arko (S4): 19.i.2015, 2 larvae; 28.ii.2015, 1 larva; 12.iii.2015, 2 larvae; 23.iv.2015, 1 larva. Tiffech Bridge (S11): 19.ii.2016, 1 larva. Oued Aar (S14): 29.i.2015, 1 larva. Aazelet Hlima (S17): 13.iii.2015, 1 larva. Chaabet Bounab (S18): 27.i.2015, 1 larva; 27.ii.2015, 1 larva. Boufrais (S19): 25.xii.2014, 1 larva; 01.i.2016, 6 larvae; 19.ii.2016, 21 larvae; 29.iii.2016, 1 larva; 23.iv.2016, 1 larvae, 3 pupae; 09.v.2016, 1 pupa. Selaoua (S20): 25.xii.2014, 1 larva.

Mesophylax aspersus is a Palearctic species. The subspecies *M. a. hoggarensis* Malicky 1998 was described from the Hoggar Mountains in the Algerian Central Sahara (Malicky 1998). *Mesophylax aspersus* was subsequently recorded in Ghoufi Canyon, in the Saharan Atlas (B.S. unpublished) and the Seybouse River (present study). Recently, Dambri *et al.* (2020) recorded it in the Aures Mountains.

Statistical analysis of the Trichoptera

The four components of the PCA that were selected represented 90.8% of total inertia. The first component (35.5% of total variance) opposed *Mesophylax aspersus* to the rest of the recorded species (Fig. 3). On only one occasion (a single larva at Oued Aar) was *M. aspersus* found in the company of other species. Its habitats were mainly upstream temporary habitats located in the semi-arid Hauts Plateaux. The second component (27.0% of total inertia) isolated two taxa, *Hydropsyche artax/lobata* and *H. gr. pellucidula*, from the two North African endemics *H. maroccana* and *H. resmineda*. The former were found to inhabit upstream fast-flowing streams and were mainly absent or rare in downstream sites. The third component (14.3% of total inertia) highlighted the dominance of *H. maroccana* at lower altitude sites while emphasizing the upstream-downstream gradient (Fig. 4). Finally, the fourth component (14.0% of total inertia) isolated *H. resmineda* which occurs downstream from *H. gr. pellucidula*.

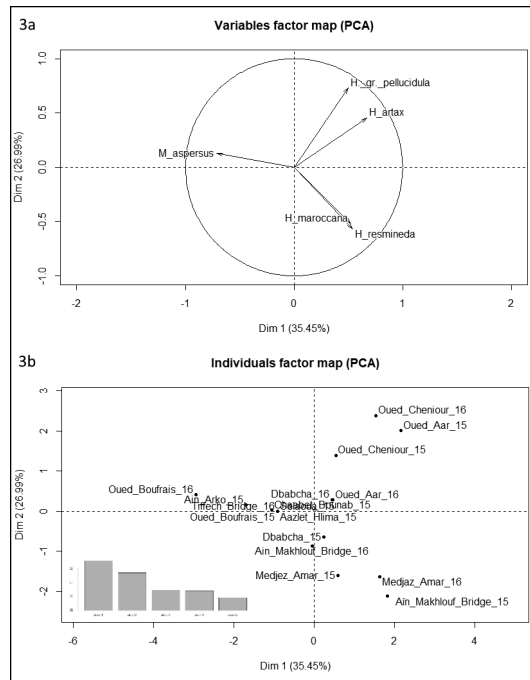


FIGURE 3: Map of principal component analysis induced by components 1x2 with screen plot (inset): 3a, species; 3b, sites.

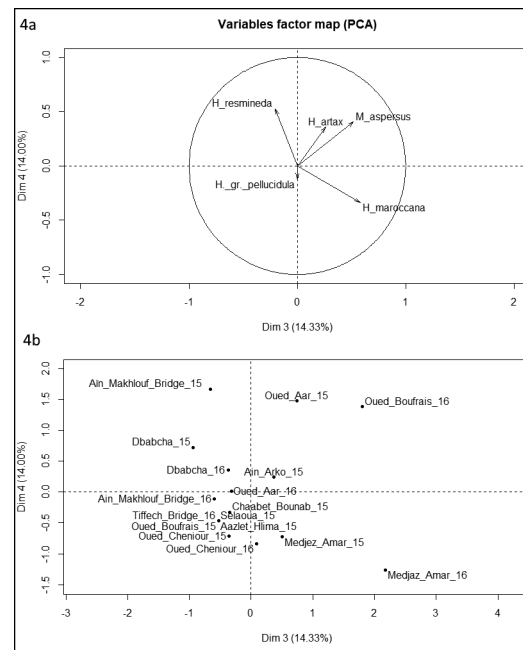


FIGURE 4: Map of principal component analysis induced by components 3x4 with scree plot (inset): 4a, species; 4b, sites.

Discussion

This is the first survey of the stoneflies and caddisflies of the Seybouse River, one of the major Algerian rivers. Three species of Plecoptera were recorded: *Capnopsis schilleri*, *Capnioneura petitpierreae*, and *Tyrrhenoleuctra tangerina*. All three species are new to the Seybouse River; also *Capnopsis schilleri* and *Capnioneura petitpierreae* are new to northeastern Algeria. All three species had been previously reported from other parts of Algeria and Khroumirie (Tunisia) (Errochdi *et al.* 2014). The larvae seem restricted to relatively moderate/high altitude temporary streams. Compared to the fauna of other Maghreb rivers, the stonefly fauna of the Seybouse River seems to be particularly impoverished. Because the sources of the Seybouse River are located in the Hauts Plateaux, the watershed lacks forested areas and dam construction has considerably reduced water flow.

Concerning the Trichoptera, *H. maroccana*, *H. resmineda* and *Mesophylax aspersus* were identified to species level whereas the taxa *H. artax/lobata* and *H. pellucidula* Species Group require further investigation. One major limitation of our study stemmed from our entire reliance on caddisfly larvae; for species determination, it is more advisable to work with pupae and adults (Waringer & Graf 2011). However, as the study area has been largely understudied, this survey provided a preliminary overview of the stoneflies and caddisflies of a major river in northeastern Algeria, thus filling important gaps. Hence, we hope that this study will stimulate further investigations based on mature pupae and adults.

Beyond their well-known role as ecological indicators (Higler & Tolkamp 1983; Houghton 2004), Hydropsychidae are widespread and often the most abundant net-spinning caddisflies. They have been dubbed “ecosystem engineers” because they perform important ecosystem services (Cardinale *et al.* 2004). In addition to filtering and recycling organic matter, they are able to stabilize gravel bed sediments and thus have a substantial effect on the physical properties of streams (Statzner *et al.* 1999; Albertson *et al.* 2019).

Hydropsychidae species have long been known to distribute themselves successively along a longitudinal gradient (Gordon & Wallace 1975; Hildrew & Edington 1979; Ruiz-García *et al.* 2012) as revealed in this study by the PCA: *H. artax/lobata* and *H. pellucidula* Species Group larvae occupied high altitude sites while *H. maroccana* was most abundant at lower altitude sites. In intermediary altitude sites, *H. resmineda* dominated the other Hydropsychidae. Our finding agrees with that of Berrahou *et al.* (2001) who showed *H. maroccana*, *H. resmineda*, and *H. lobata* as representatives of the middle stretch of the Moulouya River. This differential distribution of the Hydropsychidae may reflect their respective tolerance to a temperature gradient or other environmental factors but this hypothesis needs to be tested. One notable absence from the Seybouse River is the genus *Diplectrona* which generally inhabits headwaters (Ruiz García & Ferreras-Romero 2008).

In contrast to the hydropsychidae, *Mesophylax aspersus* occupied distinct habitats reflecting likely an adaptation to temporary streams (Ruiz-García *et al.* 2012). This hypothesis is supported by the life history of *M. aspersus* because this species is able to aestivate as adults in caves (Salavert *et al.* 2008). This behavior would allow the species to occupy intermittent streams, a common feature in the dry Mediterranean climate. The recorded larval (December–April) and pupal (April and May) phenology suggests a univoltine life cycle with larval development taking place in winter and emergence occurring in mid-spring before streams dry out. As was found in Spain (Salavert *et al.* 2008), it is likely that adults of *M. aspersus* in Algeria are troglophile during the aestival season.

Caddisflies respond differentially to environmental stressors and thus caddisfly assemblages are considered important indicators of water quality (Bowles *et al.* 2016). Their distribution is influenced also by basin geology and river zonation among other environmental factors (Bonada *et al.* 2005, 2008). A well-documented knowledge of the life history of the recorded species would help identify the drivers influencing such apparent species replacement along the upstream-downstream gradient. Some caddisflies may exhibit various adaptations (adult diapause, desiccation-resistant eggs) as found in *M. aspersus* and other macroinvertebrates (Samraoui *et al.* 1998; Agüero-Pelegrín *et al.* 1999; Samraoui 2009) while other species may have distinct affinities for temperatures, substrates, or other environmental factors.

Knowing how North African lotic macroinvertebrates react to various environmental changes, especially altered flow regimes and pollution in the context of climate change and human encroachment, may help predict species distribution and abundances. Such knowledge would further our understanding of the roles of macroinvertebrates as ecosystem engineers (Moore 2006), and will contribute towards the development of a regional biotic index, a potentially precious tool in bioassessment of North African streams and conservation measures (Lenat & Resh 2001; Bonada *et al.* 2004; Korbbaa *et al.* 2018).

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